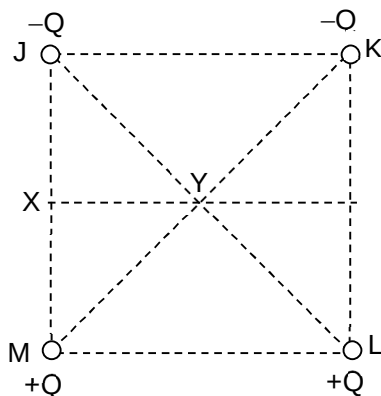


- 19 Four point charges are at the corners of a square JKLM. The point charges at J and K have a charge of $-Q$ and the point charges at L and M have a charge of $+Q$.



An electron is brought from point X to point Y in a straight line by an external force without any change in its speed. Which of the following is correct about the change in electric potential energy of the system and work done by the external force?

	change in electric potential energy	work done by the external force
A	positive	positive
B	negative	negative
C	negative	positive
D	zero	zero

The potentials at points X and Y are both zero, because they are each equidistant from the charges at J and M, and also equidistant from the charges at K and L.

Since there is no change in the electrical potential energy from X to Y, no work is done by the external force.

20 A student is given a sealed box containing a concealed electrical circuit. The student plots the